BA CAPSTONE · DATA PRODUCT · 2026

Structured Semiconductor Financial News and Volatility Dataset

for SMHandSOXX. (2020–2025)

A reusable research dataset combining 23,430 institutional news headlines with ETF market data, Garman–Klass volatility, FinBERT sentiment scores, and intraday microstructure variables

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Dataset Overview

The first openly documented dataset that aligns institutional semiconductor news with high-frequency ETF microstructure, ready for volatility forecasting and event studies.

23,430

2020–2025

1,508
TRADING DAYS

2
ETFS · SMH / SOXX

5
FORECAST HORIZONS

Use Cases

- **Financial NLP research** — Benchmarking FinBERT & LLM sentiment models
- **Volatility forecasting** — HAR, GARCH, ML & deep-learning baselines
- **Event studies** — Earnings, export-control & policy shocks
- **Semiconductor market research** — SMH / SOXX, sector cycles, capex regimes
- **AI regime analysis** — Pre / post-ChatGPT & export-control eras

08 Technical Stack

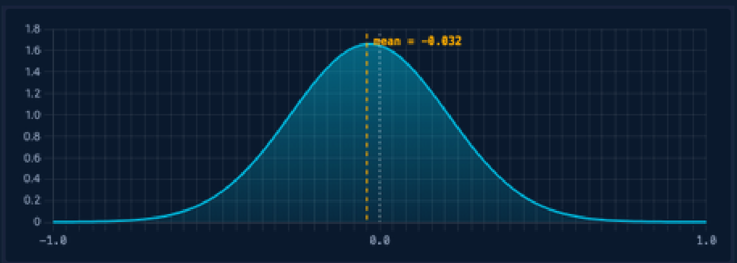
All processing in open-source Python; reproducible from the included Jupyter notebook.

Python 3.11 · pandas · numpy · yfinance · HuggingFace Transformers · FinBERT · scikit-learn · statsmodels · matplotlib · seaborn · Jupyter Notebook

09 Sentiment Distribution

FIGURE 3.1 — MONTHLY NEWS COVERAGE (ATTACHED)

DAILY NET SENTIMENT DISTRIBUTION (S_t)



Daily net sentiment S_t is approximately normal with mean −0.032 — a mild negativity bias consistent with risk-coverage emphasis in institutional headlines.